ModelIt!

Student Activity Pack

Fast Fashion Is Killing the Planet

Modeling the Environmental Cost of Cheap Clothing

Grade 09L1 | NGSS HS-ESS3-4, HS-ETS1-3 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **How many new clothing items do you think you buy in a typical month? Where do your old clothes end up?**

2. **What do you think happens to clothes after you throw them in the donation bin?**

3. **Why is fast fashion so cheap — what corners might companies be cutting to sell a T-shirt for $5?**

4. **Do you think the clothing industry has a significant environmental impact? Why or why not?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Consumer Demand | Production Rate | Textile Waste | Water Consumption | Carbon Emissions

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Business as Usual

Consumer Demand at current high levels

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Buy Less Movement

Consumer Demand reduced by 50%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Circular Fashion

High demand but with recycling and reuse systems

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Microplastic Pollution — Synthetic fabrics like polyester shed microscopic plastic fibers during washing — a single load of laundry releases up to 700,000 microplastic fibers into waterways, entering the food chain and accumulating in marine organisms and human tissue

Labor Exploitation — The social cost of fast fashion: garment workers in Bangladesh, Vietnam, and Cambodia earn as little as $2/day in unsafe conditions. The Rana Plaza collapse in 2013 killed 1,134 garment workers, exposing the human cost of cheap clothing.

Chemical Runoff — Textile dyeing and finishing use over 8,000 synthetic chemicals, many of which are discharged untreated into rivers near factories. The fashion industry is the second-largest polluter of freshwater globally, after agriculture.

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How does a $5 T-shirt end up costing the planet so much? Use evidence from your simulation to support your answers.

1. **How does your model reveal the hidden environmental cost behind a cheap price tag?**

2. **Why does reducing consumer demand have a bigger environmental impact than improving recycling?**

3. **What responsibility do consumers have versus manufacturers in reducing fashion's environmental footprint?**

4. **How does social media's influence on fashion trends connect to the environmental cascade in your model?**

5. **What would need to change about the economic system to make sustainable fashion the default?**

8 STEM Challenge: Design a Sustainable Fashion Solution

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YOUR ENGINEERING MISSION

Design a business model, technology, or system that reduces the environmental footprint of the clothing industry while remaining economically viable and appealing to teenage consumers.

THE SCENARIO

A major clothing retailer wants to become more sustainable but can't afford to lose teenage customers. They've hired your team to design a solution — whether it's a new material, a recycling system, a rental service, or a completely new business model — that dramatically reduces water, carbon, and waste while keeping clothes affordable and trendy.

Engineering Design Thinking

1. WOULD YOUR SOLUTION FOCUS ON THE PRODUCTION SIDE (MATERIALS, MANUFACTURING), THE CONSUMER SIDE (BEHAVIOR CHANGE, RENTAL MODELS), OR THE DISPOSAL SIDE (RECYCLING, UPCYCLING)?

2. HOW WOULD YOU MEASURE WHETHER YOUR SOLUTION ACTUALLY REDUCES ENVIRONMENTAL IMPACT?

3. WHAT WOULD YOUR SUSTAINABLE CLOTHING COST, AND WOULD YOUR TARGET MARKET PAY THAT PRICE?

4. HOW WOULD YOU SCALE YOUR SOLUTION TO MAKE A MEASURABLE IMPACT ON THE INDUSTRY'S 92 MILLION TONS OF ANNUAL TEXTILE WASTE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Environmental Engineers design systems to reduce industrial pollution and waste, earning $65,000–$130,000/year. Sustainable Fashion Designers create clothing using eco-friendly materials and ethical production methods, working at brands like Patagonia, Eileen Fisher, and Stella McCartney, earning $50,000–$120,000/year.